

Project Documentation: Forest Fire Prediction Web Application

This document details the configuration, files, and setup guidelines for the end-to-end Machine Learning pipeline built to predict the Fire Weather Index (FWI). The application utilizes historical climate logs from the Algerian Forest Fires dataset to run its models and delivers real-time calculation reports back to users over a lightweight web browser wrapper.

1. Repository Directory Structure

The layout of the folder structure below represents how the active workspace is organized for local development and live deployment packages:

```
testforestfires/
├── .ebextensions/
│   └── python.config
├── models/
│   ├── ridge.pkl
│   └── scaler.pkl
├── notebook/
│   ├── 2.0-EDA And FE Algerian Forest Fires.ipynb
│   ├── 3.0-Model Training.ipynb
│   └── Algerian_forest_fires_dataset_UPDATE.csv
├── templates/
│   ├── home.html
│   └── index.html
├── .gitignore
├── Algerian_forest_fires_dataset_UPDATE.csv
├── README.md
├── application.py
└── requirements.txt
```

2. AWS Configuration and Deployment Variables

The application is set up to run on AWS Elastic Beanstalk using an Amazon EC2 instance behind an nginx reverse proxy web server. Updates are pushed automatically using a linked GitHub repository connected via AWS CodePipeline.

Elastic Beanstalk Container Settings

To ensure that the cloud Python runtime container knows where to look, the configuration file inside `.ebextensions/python.config` explicitly contains the following settings layout:

```
option_settings:
  "aws:elasticbeanstalk:container:python":
    WSGIPath: application:application
```

This instruction explicitly binds the public web routing target directly to the underlying Flask instance configured inside the root level `application.py` file.

3. Analytical Feature Guide

The prediction pipeline requires the following meteorological parameters passed through the interface to perform calculation routines:

Feature Name	Analytical Description	Standard Unit / Scale
Temperature	Maximum daily recorded surface heat rating	°C
RH	Relative Atmospheric Humidity level	%
Ws	Average ambient Wind Speed readings	km/h
Rain	Total accumulated daily rainfall depth	mm
FFMC	Fine Fuel Moisture Code (Litter moisture)	Index Rating
DMC	Duff Moisture Code (Deep organic layer moisture)	Index Rating
ISI	Initial Spread Index (Expected velocity of fire spread)	Index Rating

4. Commands for Local Testing

To run the application inside a local test environment for troubleshooting or verification, use the standard command sequences listed below:

```
# Step 1. Clone the codebase locally
git clone https://github.com/Pragathithimmaiah/testforestfires.git
cd testforestfires

# Step 2. Build and trigger a safe testing environment
python3 -m venv .venv
source .venv/bin/activate

# Step 3. Install core dependencies from manifest
pip install -r requirements.txt

# Step 4. Boot up the local runtime script
python application.py
```

Once running, copy and paste the server test addresses into a browser workspace window using either `http://127.0.0.1:5000/` or internal local network shortcuts at `http://192.168.0.104:5000/`.